EASTERN MEDITERRANEAN UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING

Course Code: MENG246
Course Name: Thermodynamics II
Semester/ Year: Spring 2023-2024
Instructor: Assoc. Prof. Dr. Devrim Aydin
Assistant: Mr. Mehdi Ahmad/Mr. Hussian Shawish
Assessment: Lab
Due Date: 25 April 2024 at 19:00 hrs.

LAB NAME: Wet Cooling Tower

STUDENT(S) NO:23704019 **NAME(S), SURNAME(S): TARIK TARIQ**

SECTIONS:

No.	STUDENT OUTCOMES	COURSE LEARNING OUTCOMES	WEIGHT OF SECTION /100	MARKS OBTAINED	REMARKS
Introduction	6	7	10		
Theory	6	7	10		
Apparatus	6	7	10		
Procedure	6	7	10		
Measurements, Calculations & Results	6	7	30		
Discussion & Conclusion	6	7	30		
					Marks for report format & referencing are inclusive in the marks of each section.
TOTAL: (Out of 100)					

	Course Learning Outcomes	Student Outcomes 1	2	3	4	5	6	7
1	Understand the operating principles of gas powered thermodynamic cycles	X						
2	Understand the operating principles of vapour and combined power cycles	X						
3	Understand the operating principles of different refrigeration cycles (reversed carnot, vapour compression, absorption, advanced refrigeration cycles) and analyze their performances based on thermodynamic principles.	X						
4	Analyze the properties and p-v-t behavior of gas mixtures.	X						
5	Perform thermodynamic analysis on different processes (heating, cooling, humidification and dehumidification) by using thermodynamic relations and	X						
6	Understand the properties of different fuels and write thermodynamic	X						
7	Analyze the thermodynamic performance of wet cooling tower experimentally	X					X	
8	Analyze the thermodynamic performance of an HVAC process experimentally	X					X	
	Weight of Student Outcomes	H					M	

Instructions:

A. **Report presentation:** Draw neat, labeled diagrams where necessary. Write relevant equations and your assumptions where necessary. Be clear and specific and include units. Give explanatory notes where necessary. Carry out each step explicitly. Provide references to the data/model/equations used. References should be from the textbook or from an authentic source.

B. **Submission:** Upload your report as a pdf file to the related section in LMS within the designated time. The file uploaded should be named as **[MENG246-S2022-2023-Lab1-Student ID-Student Name].** No late submissions will be accommodated.

ABSTRACT

This experiment delves into the evaluation of a wet cooling tower's performance, crucial for managing heat in industrial processes while ensuring environmental sustainability. Employing two distinct methodologies, the study compares the outcomes of Case 1, utilizing thermodynamic principles, with those of Case 2, based on psychometric chart readings. Analysis of key parameters such as air and water flow rates reveals variations in results between the two cases, indicating differences in accuracy and precision. In Case 1, calculations derived from fundamental principles of heat and mass transfer yielded an experimental mass flow rate of air ($\dot{m}a$) of 0.5979 kg/s and a resulting water mass flow rate ($\dot{m}4$) of 0.034 kg/s. Conversely, in Case 2, readings from a psychometric chart led to an experimental $\dot{m}a$ of 2029.96 Kg/h and a theoretical $\dot{m}a$ of 2485.45 Kg/h, with an experimental water mass flow rate ($\dot{m}w4$) of 114.7 Kg/h. The discrepancies between the two cases underscore the importance of methodology selection and highlight the necessity for further investigation to refine cooling tower performance assessment, ensuring reliable and accurate data for optimizing industrial processes.

CONTENTS

LIST OF FIGURES

LIST OF GRAPHS

1. INTRODUCTION

In large scale power generation and industrial processes, managing heat efficiently is essential for operational success and environmental responsibility. Facilities like thermal power plants and nuclear reactors often need to get rid of excess heat, and sometimes, using the atmosphere as a heat sink is the only viable option due to space or environmental constraints. That's where wet cooling towers come in. These are specialized devices designed to transfer waste heat into the air around them. They work by circulating warm water from industrial processes through internal structures, allowing some of it to evaporate and take away heat from the rest of the water. This cools down the water effectively, making it suitable for reuse in industrial processes. Wet cooling towers play a crucial role not only in cooling but also in reducing environmental impacts by managing heat disposal. In this experiment, we aim to measure important factors like the flow rates of air and water at the tower's inlet and outlet, as well as humidity levels inside the tower. By doing this, we hope to better understand how these towers perform and how efficient they are in industrial settings.

2. DESCRIPTION OF APPARATUS

This cooling tower setup is like a big science kit for studying how heat and air move around in cooling towers. It's made up of different parts, like the water tank, the pump, and the fan, all working together to create a closed loop where water gets cooled down. The cool thing is you can see inside because it is made of see through material. Scientists and engineers can use it to figure out how to make cooling towers work better and use less energy. Figure 1 gives a clear picture of what wet cooling tower is composed of.

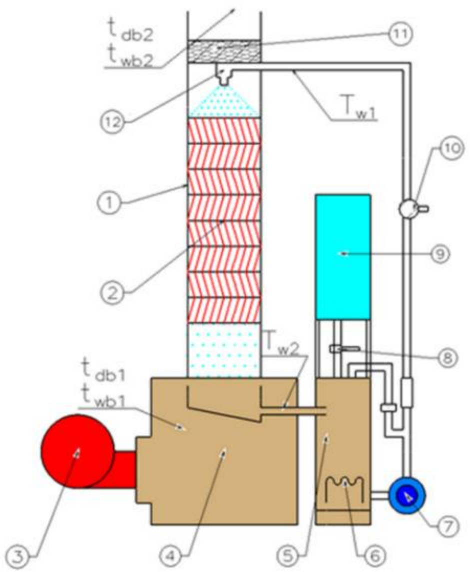

Figure 1 wet cooling tower

1. Top Column of Cooling Tower
   Water from the tank is transferred to the top column of the cooling tower.
2. Packings
   The water sprayed from the top column is uniformly distributed over the packings.

3. Compressor
   Blows air that flows among the packings inside the cooling tower.
4. Vertical Grid Apparatus (VGA) Packings
   Fill the wet cooling tower (WCT).
5. Water Tank
   Contains electrical heaters (6) that heat the water.
6. Electrical Heaters
   Heat the water in the tank.
7. Pump
   Circulates the hot water in the closed circuit.
8. Plexiglas Cooling Tower (WCT)
   Dimensions: 210 mm × 210 mm × 630 mm (L × W × H).
9. Cross-Section of Test Area
10. Types of Packings
    Made of PVC and included three types with different L/D ratios: 6, 8, and 17.
11. Radial Fan
    Generates the mass flow rate of air.
12. Measurement Devices

3. THEORY

To understand the efficiency and effectiveness of wet cooling towers, calculations often involve determining the heat transfer rates, mass balance considerations, and the impact of various operating parameters on performance.

Energy balance is given by:

$\Sigma(\dot{m}h)in = \Sigma(\dot{m}h)out$ eqn (1)

Dry air mass balance :

$\dot{m}_{a1}= \dot{m}_{a2}= \dot{m}_a$ eqn (2)

water balance

$\dot{m}_3 + \dot{m}_a w_1 = \dot{m}_4 w_2$ eqn (3)

the energy balance may be written as :

$\dot{m}_a(h_{a1} + w_1 h_{v1})+ \dot{m}_{w3}h_{w3} = \dot{m}_{a2}(h_{a2} + w_2 h_{v2}) + \dot{m}_{w4}h_{w4}$ eqn (4)

For atmospheric air, the enthalpy of the water vapor h_v may be approximate by h_g at the given temperature and for the water stream enthalpy h_w may be evaluated by h_f at the given temperature. Also recall that $h_a = c_{p\text{-}air}T$

$\dot{m}a1(Cp(T1\text{-}T2) + \omega 1\ (hg1 – hf4) + \omega 2\ (hg2 – hf3\ ) = \dot{m}w3\ (hf4 – hf3)$ eqn (5)

In many uses of cooling towers the value of m_{w3} is known, as well as the temperature drop needed for the water stream. The above equation is then used to calculate the mass flow rate of air required on a dry basis. The mass flow rate of dry air may also be calculated directly from

Eq. (1) if data from a psychometric chart are to be used. The result is:

$\dot{m}a = \dot{m}w3\ (hf4 - hf3) / [(hm1 - hm2) - (\omega 2 - \omega 1)hf4]$ eqn (6)

4. EXPERIMENTAL PRECEDURE

To experimentally operate the wet cooling tower located at the university premises, firstly, I would initiate the compressor and water heater. Subsequently, I'd open the valve for water flow, allowing it to gradually fill the system from the top to the bottom stage. To accurately assess the system's performance, including parameters such as wet bulb temperature (Twb), dry bulb temperature (Tdb), and the rates of water and air, a waiting period of approximately five minutes would be necessary. Adhering to safety guidelines outlined in the provided handout, it's essential to prevent the inlet temperature from exceeding 50 degrees Celsius. Following these procedures, the water would commence the chilling process, completing the experimental run.

Table 1 THE RECORDED EPERIMENTAL DATA

Pressure (KPa)	Air inlet (Temp) $^\circ C$		Air outlet $^\circ C$		Water inlet C°	Water outlet $^\circ C$	Constant values		
	(wb1)	(db1)	(wb2)	(db2)	(T-w3)	(T-w4)	(m_w3) Kg/h	(m_w4) Kg/h	(V_air) m^3/h
97	17	20	20	25	65	25	120	47	450

5. CALCULATIONS AND RESULTS

PART A

For the inlet air:

$\omega_2 = (0.622\ P_v)/(P-P_v)$

Pv = P sat @ 17 °C = 1.9591 kpa

ω_2 = (0.622*1.9591 kpa)/(97-1.9591 kpa) = **0.01282 kg water/kg dry air**

ω_1 = (Cp(T2 - T1) + ω2 * hfg) / (hg1 - hf1)

ω_1 = (1.005*(17 - 20) + (0.01282 * 2460.64) / (2531.94 – 71.3552) =

0.01159 kg water/kg dry air

$h_1 = CpT_1 + \omega_1 hg_1$ = 1.005*20 + 0.01159 *2537.4 = **49.5 kJ/kg**

for the outlet

$h_3 = h_{f@65\ ^\circ C}$ = **272.12 kj/kg**

$h_4 = h_{f@25\ ^\circ C}$ = **104.83 kj/kg**

$\omega_2 = (0.622\ P_v)/(P-P_v)$

Pv = P sat @ 17 °C = 2.3392 kpa

ω_2 = (0.622*2.3392 kpa)/(97-2.3392 kpa) = **0.01537 kg water/kg dry air**

$\omega_1 = (Cp(T2 - T1) + \omega 2 * h_{fg2}) / (hg1 - hf1)$
$\omega_1 = (1.005*(20 - 25) + (0.01537 * 2453.5) / (2546.5 – 83.915) =$
0.01327 kg water/kg dry air
$h_2 = CpT_1 + \omega_1 hg_1 = 1.005*25 + 0.01327 *2546.5 =$ **58.91 kJ/kg**

by using the mass water balance equation we can obtain $\dot{m}_4$:

$\dot{m}_a[Cp\ air\ (T1 - T2) + \omega 1\ (h_{g1} - h_{f4}) - \omega 2(hg2 - hf4)] = mw3(hf4 - hf3)$

$\dot{m}_{a\ theoretical}$= 0.033(104.83 – 272.12)/ [1.005 (17 - 20) + 0.01159 (2537.4 – 104.83) – 0.01327 (2546.5 – 104.83)] =
0.764kg/s

$\dot{m}_{a\ experimental}$ = 0.033(272.12 – 104.83)/[(58.91 kJ/kg -49.5)-(0.01327 - 0.01159)104.83] = **0.5979 kg/s**

$\dot{m}_4 = m_3 + m_a(\omega_1 - \omega_2)$ = 0.033 + 0.5979*(0.01327 - 0.01159) = **0.034kg/s = 122.4 kg/h**

PART B

Using psychometric chart to read enthalpies and absolute humidity. Then we will use the values obtained from the psychometric chart to calculate the mass flow rate of dry air (ma) as well as $\dot{m}_4$.

Inlet

ω_1 = 10.75/1000 = 0.01075 kg v /kg a

h1 = 48kj/kg

Outlet

ω_2 = 0.01275 kg v /kg a

h_2= 58kj/kg

Obtaining $\dot{m}_4$:

$\dot{m}_4 = m_3 + m_a(\omega_1 - \omega_2)$,

To calculate for $\dot{m}_{a(experimental)}$

$\dot{m}_a = [\dot{m}_{w3}(h_{f3}- h_{f4})] / [(h_{m2}-h_{m1}) - (\omega_2-\omega_1) h_{f4}]$

$\dot{m}_a$ = [0.033kg/s (**272.12** – 104.83)] / [(58 – 48) – (0.01275 – 0.01075) x 104.83 **]**

$\dot{m}_{a(experimental)} = 2029.96 \frac{Kg}{h}$

To calculate for $\dot{m}_{a(theoretical)}$

$\dot{m}_a [C_{p\text{-}air}(T_1\text{-}T_2) + \omega_1(h_{g1}\text{-}h_{f4})\text{-} \omega_2(h_{g2}\text{-}h_{f4})] = \dot{m}_{w3}(h_{f4}\text{-} h_{f3})$

$\dot{m}_{a\ theoretical}$= 0.033(104.83 – 272.12)/ [1.005 (17 - 20) + 0.01075 (2537.4 – 104.83) – 0.01275 (2546.5 – 104.83)] = **0.6904kg/s**

$\boldsymbol{\dot{m}_{a(theoretical)} = 2485.45\ \frac{Kg}{h}}$

To calculate for $\dot{m}_{w4(experimental)}$

$\dot{m}_{w4} = \dot{m}_{w3} + \dot{m}_a\omega_1 - \dot{m}_a\omega_2$

$\dot{m}_{w4} = 0.033\frac{Kg}{s} + 0.5639\frac{Kg}{s}\ (0.01075\ \frac{Kg\ water}{Kg\ Air}) – 0.5639\frac{Kg}{h}\ (0.01275\frac{Kg\ water}{Kg\ Air})$

$\boldsymbol{\dot{m}_{w4(exp)} = 0.03187\ \frac{Kg}{s}\ = 114.7\ \frac{Kg}{h}}$

Table 2 CASE 1 AND 2 EXPERIMENTAL AND THEORETICAL COMPARISON

#	ω_1	ω_2	h_{m1}	h_{m2}	$\dot{m}_{w4(exp)}$	$\dot{m}_{w4(thy)}$	$\dot{m}_{a(exp)}$	$\dot{m}_{a(thy)}$
Case-1; Using formulas	0.01159 $\frac{Kg\ water}{Kg\ Air}$	0.01327 $\frac{Kg\ water}{Kg\ Air}$	49.5 $\frac{KJ}{Kg}dry\ air$	58.91 $\frac{KJ}{Kg}dry\ air$	$122.4\frac{Kg}{h}$	$47\frac{Kg}{h}$	2152.44 $\frac{Kg}{h}$	2750.4 $\frac{Kg}{h}$
Case-2; Using psychometric chart	0.01075 $\frac{Kg\ water}{kg\ air}$	0.01275 $\frac{Kg\ water}{kg\ air}$	48 $\frac{KJ}{kg}$ dry air	58 $\frac{KJ}{kg}$ dry air	114.7 $\frac{Kg}{h}$	$47\frac{Kg}{h}$	2029.96 $\frac{Kg}{h}$	2485.45 $\frac{Kg}{h}$

6. DISCUSSION OF RESULTS

The experimental procedure outlined for operating the wet cooling tower involved initiating the compressor and water heater, followed by gradually allowing water to fill the system while monitoring parameters such as wet bulb temperature, dry bulb temperature, and flow rates. The subsequent calculations and results from two parts of the experiment revealed insights into the performance of the cooling tower. In Part A, calculations based on thermodynamic principles provided values for air and water properties at the inlet and outlet of the tower, leading to an experimental mass flow rate of air (ṁa) of 0.5979 kg/s and a resulting water mass flow rate (ṁ4) of 0.034 kg/s. Part B utilized psychometric chart readings to determine air and water properties, yielding an experimental ṁa of 2029.96 Kg/h and a theoretical ṁa of 2485.45 Kg/h, with an experimental ṁw4 of 114.7 Kg/h. These results collectively underscore the importance of both theoretical calculations and experimental validation in understanding the performance characteristics of cooling tower systems, highlighting areas for potential optimization and further investigation.

7. CONCLUSION

Part A and Part B of the experiment utilized different methods for calculating important parameters related to the performance of the wet cooling tower. In Part A, thermodynamic principles were applied to determine air and water properties at the inlet and outlet of the tower, leading to the calculation of mass flow rates of air and water. These calculations were based on equations derived from fundamental principles of heat and mass transfer. On the other hand, Part B utilized readings from a psychometric chart to determine air and water properties, which were then used to calculate mass flow rates. This approach relied on graphical interpretations of air and water properties and their relationships, providing an alternative method to determine the same parameters. Consequently, differences in the results between Part A and Part B may arise due to variations in calculation methods, accuracy of measurements, and assumptions made in each approach. Despite these differences, both methods contribute valuable insights into the performance of cooling tower systems, offering complementary perspectives for understanding and optimizing their operation.

8. REFERENCES

ASHRAE-psychrometric-chart-IP-2-_dry-bulb-temperature-600. (2020, December 3). Energy Vanguard. https://www.energyvanguard.com/blog/fundamentals-of-psychrometrics-part-3/attachment/attachment-ashrae-psychrometric-chart-ip-2-_dry-bulb-temperature-600/

Çengel, Y. A., Boles, M. A., & Kanoglu, M. (2018). *Thermodynamics: An engineering approach*.

Yunus A. Cengel, D., & Boles, M. A. (2018). *Property tables booklet for thermodynamics: An engineering approach*. McGraw-Hill Education.

9. APPENDIX

The figure below is a psychometric chart that is used to locate the required data for the calculation section. The yellow lines represent outlet data while the red lines represents inlet data.

A

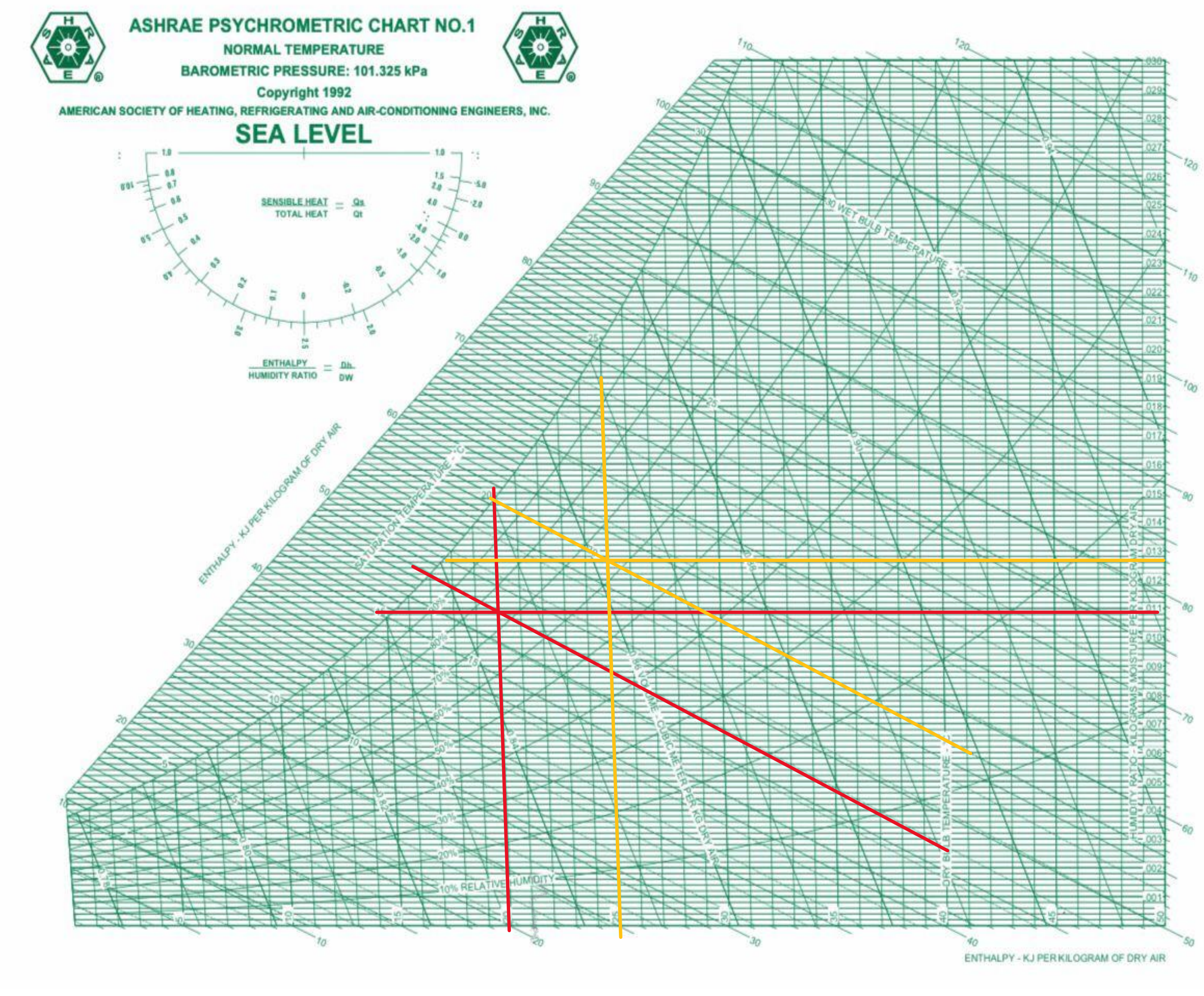

Figure 2 psychometric chart